

	"Empowerment through quality technical education" AJEENKYA DY Patil School of Engineering Dr. D. Y. Patil Knowledge City, Charholi Bk., Via. Lohegaon, Pune-412105. Department of Artificial Intelligence and Data Science Engineering	
ACADEMIC YEAR : 2025-26		
SUBJECT: Database Management Systems (PCC-251- AID)		
CLASS: SE-B	SEMESTER: IV	
TITLE OF MINI PROJECT: Design & Implementation of a Hospital Management System Using SQL		
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TITLE OF MINI PROJECT - Hospital Management System Using MySQL with Automated Billing

Introduction-

A Hospital Management System (HMS) is essential for maintaining accurate and efficient records of patients, doctors, appointments, treatments, and billing. This project demonstrates how database concepts such as tables, constraints, joins, triggers, and stored procedures can be used to build a functional and automated hospital database system. The system ensures faster data retrieval, reduced manual errors, and improved workflow in hospital operations.

Problem Statement –

Hospitals handle large volumes of data daily, including patient details, doctor schedules, appointments, treatments, and billing. Manual record-keeping often leads to errors, delays, and difficulty in retrieving information. There is a need for a structured and automated system that can store, manage, and process hospital data efficiently. This project aims to design and implement a Hospital Management System using MySQL that automates billing, maintains accurate records, and provides quick access to essential information.

Implementation -

Include:

1. Database Creation

- Creation of database HOSPITAL_RECORD

2. Table Creation

- Patient
- Doctor
- Appointment
- Treatment
- Bill

3. Data Insertion

- 5 sample patients
- 5 sample doctors
- 5 appointments
- 5 treatments

4. Trigger

- trg_generate_bill to auto-generate bills when appointment status becomes *Completed*

5. Stored Procedure

- GetPatientDetails to fetch patient + appointment details

6. SQL Queries

- Joins
- Aggregation
- Reports

□ Input/output snapshots (screenshots or console output)

```
mysql> USE HOSPITAL_RECORD;
Database changed
mysql> INSERT INTO Patient (PatientID, Name, Age, Gender, Phone, Address)
-> VALUES
-> (1, 'Amit Sharma', 32, 'Male', '9876543210', 'Pune'),
-> (2, 'Priya Deshmukh', 27, 'Female', '9823456712', 'Mumbai'),
-> (3, 'Rohan Patil', 45, 'Male', '9867543210', 'Nashik'),
-> (4, 'Sneha Mulkarni', 30, 'Female', '9812345678', 'Nagpur'),
-> (5, 'Vikas Jadhav', 52, 'Male', '9898765432', 'Aurangabad');
ERROR 1062 (23000): Duplicate entry '1' for key 'patient.PRIMARY'
mysql> select * from Patient;
+-----+-----+-----+-----+-----+-----+
| PatientID | Name       | Age | Gender | Phone       | Address |
+-----+-----+-----+-----+-----+-----+
| 1         | Amit Sharma | 32  | Male   | 9876543210 | Pune    |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.01 sec)

mysql> INSERT INTO Patient (PatientID, Name, Age, Gender, Phone, Address)
-> VALUES
-> (2, 'Priya Deshmukh', 27, 'Female', '9823456712', 'Mumbai'),
-> (3, 'Rohan Patil', 45, 'Male', '9867543210', 'Nashik'),
-> (4, 'Sneha Mulkarni', 30, 'Female', '9812345678', 'Nagpur'),
-> (5, 'Vikas Jadhav', 52, 'Male', '9898765432', 'Aurangabad');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '-> VALUES
(2, 'Priya Deshmukh', 27, 'Female', '9823456712', 'Mumbai'),
-> {' at line 2
mysql> INSERT INTO Patient Values(2, 'Priya Deshmukh', 27, 'Female', '9823456712', 'Mumbai');
Query OK, 1 row affected (0.08 sec)

mysql> INSERT INTO Patient Values(3, 'Rohan Patil', 45, 'Male', '9867543210', 'Nashik');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Patient Values(4, 'Sneha Mulkarni', 30, 'Female', '9812345678', 'Nagpur');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Patient Values(5, 'Vikas Jadhav', 52, 'Male', '9898765432', 'Aurangabad');
Query OK, 1 row affected (0.07 sec)
```

```
mysql> INSERT INTO Doctor (DoctorID, Name, Specialization, Phone, Fees)
-> VALUES (1, 'Dr. Meera Joshi', 'Cardiologist', '9876581234', 800);
Query OK, 1 row affected (0.09 sec)

mysql> INSERT INTO Doctor (DoctorID, Name, Specialization, Phone, Fees)
-> VALUES (2, 'Dr. Rajesh Mulkarni', 'Orthopedic', '9823459887', 600);
Query OK, 1 row affected (0.05 sec)

mysql> INSERT INTO Doctor (DoctorID, Name, Specialization, Phone, Fees)
-> VALUES (3, 'Dr. Sneha Patil', 'Dermatologist', '9812345678', 500);
Query OK, 1 row affected (0.07 sec)

mysql> INSERT INTO Doctor (DoctorID, Name, Specialization, Phone, Fees)
-> VALUES (4, 'Dr. Amit Deshmukh', 'Neurologist', '9898765432', 900);
Query OK, 1 row affected (0.07 sec)

mysql> INSERT INTO Doctor (DoctorID, Name, Specialization, Phone, Fees)
-> VALUES (5, 'Dr. Priya Shah', 'Pediatrician', '9867543219', 700);
Query OK, 1 row affected (0.07 sec)

mysql> SELECT * FROM Doctor;
+-----+-----+-----+-----+-----+
| DoctorID | Name       | Specialization | Phone      | Fees  |
+-----+-----+-----+-----+-----+
| 1        | Dr. Meera Joshi | Cardiologist   | 9876581234 | 800.00 |
| 2        | Dr. Rajesh Mulkarni | Orthopedic    | 9823459887 | 600.00 |
| 3        | Dr. Sneha Patil | Dermatologist  | 9812345678 | 500.00 |
| 4        | Dr. Amit Deshmukh | Neurologist    | 9898765432 | 900.00 |
| 5        | Dr. Priya Shah | Pediatrician   | 9867543219 | 700.00 |
+-----+-----+-----+-----+-----+
5 rows in set (0.03 sec)

mysql> SELECT * FROM Patients;
ERROR 1146 (42S02): Table 'hospital_record.patients' doesn't exist
mysql> SELECT * FROM Patient;
+-----+-----+-----+-----+-----+-----+
| PatientID | Name       | Age | Gender | Phone      | Address |
+-----+-----+-----+-----+-----+-----+
| 1         | Amit Sharma | 32  | Male   | 9876543218 | Pune    |
+-----+-----+-----+-----+-----+-----+
```

```
5 rows in set (0.03 sec)

mysql> SELECT * FROM Patients;
ERROR 1146 (42S02): Table 'hospital_record.patients' doesn't exist
mysql> SELECT * FROM Patient;
+-----+-----+-----+-----+-----+-----+
| PatientID | Name       | Age | Gender | Phone      | Address |
+-----+-----+-----+-----+-----+-----+
| 1         | Amit Sharma | 32  | Male   | 9876543218 | Pune    |
| 2         | Priya Deshmukh | 27  | Female | 9823456712 | Mumbai |
| 3         | Rohan Patil | 45  | Male   | 9867543210 | Nashik  |
| 4         | Sneha Mulkarni | 38  | Female | 9812345678 | Nagpur  |
| 5         | Vikas Jadhav | 52  | Male   | 9898765432 | Aurangabad |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> INSERT INTO Appointment (ApptID, PatientID, DoctorID, ApptDate, Status)
-> VALUES (1, 1, 1, '2026-04-27', 'Scheduled');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Appointment (ApptID, PatientID, DoctorID, ApptDate, Status)
-> VALUES (2, 2, 3, '2026-04-28', 'Scheduled');
Query OK, 1 row affected (0.04 sec)

mysql> INSERT INTO Appointment (ApptID, PatientID, DoctorID, ApptDate, Status)
-> VALUES (3, 3, 2, '2026-04-29', 'Pending');
Query OK, 1 row affected (0.07 sec)

mysql> INSERT INTO Appointment (ApptID, PatientID, DoctorID, ApptDate, Status)
-> VALUES (4, 4, 5, '2026-04-30', 'Scheduled');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Appointment (ApptID, PatientID, DoctorID, ApptDate, Status)
-> VALUES (5, 5, 4, '2026-05-01', 'Scheduled');
Query OK, 1 row affected (0.07 sec)

mysql> SELECT * FROM Appointment;
+-----+-----+-----+-----+-----+
| ApptID | PatientID | DoctorID | ApptDate | Status |
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
```

```

mysql> SELECT * FROM Appointment;
+-----+-----+-----+-----+-----+
| ApptID | PatientID | DoctorID | ApptDate | Status |
+-----+-----+-----+-----+-----+
| 1 | 1 | 1 | 2026-04-27 | Scheduled |
| 2 | 2 | 3 | 2026-04-28 | Scheduled |
| 3 | 3 | 2 | 2026-04-29 | Pending |
| 4 | 4 | 5 | 2026-04-30 | Scheduled |
| 5 | 5 | 4 | 2026-05-01 | Scheduled |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> INSERT INTO Treatment (TreatmentID, ApptID, Diagnosis, Prescription, Notes)
-> VALUES (1, 1, 'High Blood Pressure', 'Amlodipine 5mg daily', 'Monitor BP daily');
Query OK, 1 row affected (0.07 sec)

mysql> INSERT INTO Treatment (TreatmentID, ApptID, Diagnosis, Prescription, Notes)
-> VALUES (2, 2, 'Skin Allergy', 'Cetirizine 10mg', 'Avoid dust exposure');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Treatment (TreatmentID, ApptID, Diagnosis, Prescription, Notes)
-> VALUES (3, 3, 'Knee Pain', 'Ibuprofen 400mg', 'Apply ice pack twice daily');
Query OK, 1 row affected (0.07 sec)

mysql> INSERT INTO Treatment (TreatmentID, ApptID, Diagnosis, Prescription, Notes)
-> VALUES (4, 4, 'Fever & Cold', 'Paracetamol 650mg', 'Drink warm fluids');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Treatment (TreatmentID, ApptID, Diagnosis, Prescription, Notes)
-> VALUES (5, 5, 'Migraine', 'Sumatriptan 50mg', 'Avoid bright light');
Query OK, 1 row affected (0.07 sec)

mysql> SELECT * FROM Treatment;
+-----+-----+-----+-----+-----+
| TreatmentID | ApptID | Diagnosis | Prescription | Notes |
+-----+-----+-----+-----+-----+
| 1 | 1 | High Blood Pressure | Amlodipine 5mg daily | Monitor BP daily |
| 2 | 2 | Skin Allergy | Cetirizine 10mg | Avoid dust exposure |
| 3 | 3 | Knee Pain | Ibuprofen 400mg | Apply ice pack twice daily |
| 4 | 4 | Fever & Cold | Paracetamol 650mg | Drink warm fluids |
+-----+-----+-----+-----+-----+

```

```

mysql> ALTER TABLE Bill
MODIFY BillID INT AUTO_INCREMENT;
mysql> ALTER TABLE Bill
-> MODIFY BillID INT AUTO_INCREMENT;
Query OK, 0 rows affected (0.17 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_generate_bill
-> AFTER UPDATE ON Appointment
-> FOR EACH ROW
-> BEGIN
-> IF NEW.Status = 'Completed' AND OLD.Status <> 'Completed' THEN
-> INSERT INTO Bill (PatientID, Amount, BillDate, PaymentStatus)
-> VALUES (NEW.PatientID, 500, CURDATE(), 'Pending');
-> END IF;
-> END $$
Query OK, 0 rows affected (0.07 sec)

mysql> DELIMITER ;
mysql> UPDATE Appointment
-> SET Status = 'Pending'
-> WHERE ApptID = 1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Appointment
-> SET Status = 'Completed'
-> WHERE ApptID = 1;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM Bill;
+-----+-----+-----+-----+-----+
| BillID | PatientID | Amount | BillDate | PaymentStatus |
+-----+-----+-----+-----+-----+
| 1 | 1 | 500.00 | 2026-04-27 | Pending |
+-----+-----+-----+-----+-----+

```

```

1 row in set (0.00 sec)

mysql> ALTER TABLE Bill
-> MODIFY BillID INT AUTO_INCREMENT;
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_generate_bill
-> AFTER UPDATE ON Appointment
-> FOR EACH ROW
-> BEGIN
-> IF NEW.Status = 'Completed' AND OLD.Status <> 'Completed' THEN
-> INSERT INTO Bill (PatientID, Amount, BillDate, PaymentStatus)
-> VALUES (NEW.PatientID, 500, CURDATE(), 'Pending');
-> END IF;
-> END $$
ERROR 1359 (HY000): Trigger already exists

mysql> DELIMITER ;
mysql> SELECT p.PatientID, p.Name AS PatientName, d.Name AS DoctorName, d.Specialization
-> FROM Appointment a
-> JOIN Patient p ON a.PatientID = p.PatientID
-> JOIN Doctor d ON a.DoctorID = d.DoctorID;

```

PatientID	PatientName	DoctorName	Specialization
1	Amit Sharma	Dr. Meera Joshi	Cardiologist
2	Priya Deshmukh	Dr. Sneha Patil	Dermatologist
3	Rohan Patil	Dr. Rajesh Mulkarni	Orthopedic
4	Sneha Mulkarni	Dr. Priya Shah	Pediatrician
5	Vikas Jadhav	Dr. Amit Deshmukh	Neurologist

```

5 rows in set (0.02 sec)

mysql> SELECT a.ApptID, p.Name AS Patient, d.Name AS Doctor,
-> t.Diagnosis, t.Prescription
-> FROM Appointment a
-> JOIN Patient p ON a.PatientID = p.PatientID
-> JOIN Doctor d ON a.DoctorID = d.DoctorID
-> JOIN Treatment t ON a.ApptID = t.ApptID;

```

```

mysql> SELECT a.ApptID, p.Name AS Patient, d.Name AS Doctor,
-> t.Diagnosis, t.Prescription
-> FROM Appointment a
-> JOIN Patient p ON a.PatientID = p.PatientID
-> JOIN Doctor d ON a.DoctorID = d.DoctorID
-> JOIN Treatment t ON a.ApptID = t.ApptID;

```

ApptID	Patient	Doctor	Diagnosis	Prescription
1	Amit Sharma	Dr. Meera Joshi	High Blood Pressure	Amlodipine 5mg daily
2	Priya Deshmukh	Dr. Sneha Patil	Skin Allergy	Cetirizine 10mg
3	Rohan Patil	Dr. Rajesh Mulkarni	Knee Pain	Ibuprofen 400mg
4	Sneha Mulkarni	Dr. Priya Shah	Fever & Cold	Paracetamol 650mg
5	Vikas Jadhav	Dr. Amit Deshmukh	Migraine	Sumatriptan 50mg

```

5 rows in set (0.00 sec)

mysql> SELECT b.BillID, p.Name AS Patient, b.Amount, b.BillDate, b.PaymentStatus
-> FROM Bill b
-> JOIN Patient p ON b.PatientID = p.PatientID;

```

BillID	Patient	Amount	BillDate	PaymentStatus
1	Amit Sharma	500.00	2026-04-27	Pending

```

1 row in set (0.00 sec)

mysql> SELECT COUNT(*) AS TotalPatients FROM Patient;

```

TotalPatients
5

```

1 row in set (0.00 sec)

mysql> SELECT Status, COUNT(*) AS Total
-> FROM Appointment
-> GROUP BY Status;

```

```
mysql> SELECT b.BillID, p.Name AS Patient, b.Amount, b.BillDate, b.PaymentStatus
-> FROM Bill b
-> JOIN Patient p ON b.PatientID = p.PatientID;
+-----+-----+-----+-----+-----+
| BillID | Patient | Amount | BillDate | PaymentStatus |
+-----+-----+-----+-----+-----+
| 1 | Amit Sharma | 500.00 | 2026-04-27 | Pending |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) AS TotalPatients FROM Patient;
+-----+
| TotalPatients |
+-----+
| 5 |
+-----+
1 row in set (0.06 sec)

mysql> SELECT Status, COUNT(*) AS Total
-> FROM Appointment
-> GROUP BY Status;
+-----+-----+
| Status | Total |
+-----+-----+
| Completed | 1 |
| Scheduled | 3 |
| Pending | 1 |
+-----+-----+
3 rows in set (0.05 sec)

mysql>
```

Advantages and Limitations-

Advantages

- Automates billing using triggers
- Reduces manual errors
- Fast and accurate data retrieval
- Easy to maintain and update
- Ensures data integrity using constraints
- Real-world hospital workflow simulation

Limitations

- No frontend interface (CLI only)
- Limited to basic hospital operations
- No authentication or user roles
- No advanced analytics

Future Scope-

- Add a web or mobile frontend
- Implement login system with roles (Admin, Doctor, Receptionist)
- Add NoSQL (MongoDB) for unstructured data

- Integrate payment gateway
- Add appointment reminders via SMS/email
- Add dashboards for analytics

Conclusion-

The Hospital Management System successfully demonstrates how MySQL can be used to design and implement a structured, automated, and efficient database system. Using triggers, stored procedures, and SQL queries, the system reduces manual effort, improves accuracy, and provides quick access to essential hospital information. This project highlights the importance of database management in real-world applications.

References-

- MySQL Documentation
- DBMS Textbooks
- Online SQL Tutorials
- Classroom Notes